minimize execution costs. Further, the Commission understands that competing with the most effective SORs is more difficult without possessing real-time market data while minimizing data latency,[1810] and that those traders who do not access trading tools that utilize comprehensive market data with low-latency experience higher execution costs on average.

One commenter stated that the association of broker-dealers with proprietary data feeds represents behavior that is "largely window-dressing," and that broker-dealers still rely heavily on the SIP.[1811] This commenter also stated that many broker-dealers have "…layers of market data normalization and aggregation by third-party vendors" which further increase the latency of the data as it is used.[1812] To the extent this it is the case that any current subscriber of proprietary data feeds does not, in fact, make good use of them according to the most competitive standards, the Commission believes it represents a further way in which more capable users of market data are separated from less capable users of market data, and not an indication that proprietary market data feeds are of no real advantage to any broker-dealer.

C. Economic Effects of the Rule

1. Consolidated Market Data

The Commission believes that the enhancements to consolidated data will result in numerous beneficial economic effects. These economic effects derive from codifying the

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[1810] Some commenters supported the idea that proprietary data is important in order to be competitive in offering executing brokerage services. See, e.g., Healthy Markets Letter I at 5; T. Rowe Price Letter at 1.

[1811] See BestEx Research Letter at 3.

[1812] See BestEx Research Letter at 4.

definition of core data, from redefining the round lot, and from expanding the content of core data.

The change will have the benefit of mitigating the influence of existing conflicts of interest inherent in the existing exclusive SIP model.[1813] The change reduces the divergence between exchanges' proprietary DOB products and current SIP data because it establishes data elements that competing consolidators can include in their consolidated market data products.

One commenter stated that the claim that the expansion of core data mitigates conflicts of interest fails to consider the fact that the Governance Order gives some non-SRO market participants voting power on the effective national market system plan(s) Operating Committee.[1814] This commenter stated that the non-SROs would have a conflict of interest and that this needs to be considered when discussing any conflicts mitigated by the rules. The Commission disagrees with this commenter's assessment. It is not clear how the introduction of non-SRO votes to the Operating Committee and their associated interests are relevant to the question of whether or not the expansion of core data will mitigate the conflicts of interests of the SRO members of the Operating Committee. As discussed above,[1815] the Governance Order will reduce, but not eliminate the conflicts of interest of the SROs on the Operating Committee. The potential for further mitigation of the influence of those conflicts remains and the Commission continues to believe that the expansion of core data will have that benefit.

1813 See supra Section V.A.2. for a discussion of these conflicts of interest.

1814 See Nasdaq Letter IV at 49.

1815 See supra Section III.I.

(a) Definitions of Consolidated Market Data, Core Data, Administrative Data, and Regulatory Data

The Commission's definitions of "consolidated market data," "consolidated market data product," "core data," "regulatory data," "administrative data," and "self-regulatory organization-specific program data" under Regulation NMS will specify the quotation and transaction information in NMS stocks that can be collected, consolidated, and disseminated under rules of the national market system and pursuant to an effective national market system plan(s). This definition will codify the dissemination of certain current SIP data elements, and will include some additional data elements, but will not include some data that the exclusive SIPs currently disseminate. This section discusses the secondary economic effects of this expansion to consolidated market data that will come from codifying the inclusion of some current SIP data in "consolidated market data," while subsequent sections discuss the economic effects of the new round lot definition and expanding the content of core data. These secondary effects are providing flexibility to the Equity Data Plans for including new data elements, requiring that regulatory data will continue to be provided in the decentralized consolidation model, cost to update the national market system plan(s), and costs to obtain data that is currently in SIP data but will not be included in consolidated market data, such as data on information related to OTC equities, certain corporate bonds, and indices.[1816]

The Commission believes the definitions of "self-regulatory organization-specific program data," "regulatory data" and "administrative data," along with the ability for the Equity Data Plan(s) to add elements to these proposed definitions, will promote regulatory efficiency by providing flexibility for consolidated market data to include data elements beyond those

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[1816] See supra Section II.C.2(c).

explicitly defined as "consolidated market data." It provides a mechanism for the participants in the national market system plan(s) to propose to add additional data elements, such as elements similar to current retail liquidity programs. This will allow for organic change in consolidated market data that may become useful due to future market and regulatory developments. Further, while the underlying data content of "regulatory data" is currently included in disseminated SIP data, the definition of "regulatory data" will help ensure that market participants continue to have access to this information as part of consolidated market data.[1817]

The Commission recognizes that the Equity Data Plan(s) will incur one-time initial implementation costs in ensuring the plans are consistent with the proposed definitions of "consolidated market data," "core data," "administrative data," "regulatory data," and "self-regulatory organization-specific program data," but the plans will not incur significant ongoing costs as a result of the codification of these five definitions.[1818] These initial implementation costs will come from the Operating Committees needing to draft revisions to their respective plans that are consistent with the proposed definitions.

The Commission believes that not including some data elements that the exclusive SIPs currently transmit[1819] in the definition of "consolidated market data" may have some costs to

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[1817] Commenters agreed that regulatory data is highly relevant and important to all types of market participants. See, e.g., IEX Letter at 7; MEMX Letter at 6.

[1818] See infra Section V.C.2(d)(ii) and supra Section IV.D.6(c) for a discussion of these costs. Below in Section V.C.1(c)(iv), the Commission also discusses the costs of including data elements to the definition of "core data" that are not currently in SIP data.

[1819] See supra Section II.C.2(c) and Proposing Release, 85 FR at Section III.B.

those market participants who would want to arrange to get this data elsewhere.[1820] The UTP SIP offers quotation and transaction feeds for OTC equities, and the CTA Plan permits the dissemination of "concurrent use" data related to corporate bonds and indexes.[1821] Under the amendments, these data elements will not be defined as consolidated market data or core data elements. However, the amendments will not preclude the provision of these data elements by the SROs via proprietary data products to market participants and investors who wish to receive them.

One commenter stated that not including quotation and transaction data for OTC equities in consolidated market data may reduce market participant access to this data and would increase both the costs to the SRO to provide the data and the costs of market participants to acquire it.[1822] This commenter also stated that, because OTC equities may become listed and become NMS stocks and vice versa, not providing this data in the same feed as core data could result in a disruption of market data when a security switches between being listed and unlisted and investors or market participants are not subscribed to both services providing core data and data for delisted issuers.[1823] The Commission acknowledges that not including information related to OTC equities in consolidated market data may potentially increase the costs of FINRA providing this data and market participants to acquire the data. The Commission also acknowledges that

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1820 One commenter agreed, stating that not including quotation and transaction data for OTC equities in consolidated market data would increase both the costs to provide the data and the costs of market participants to acquire it. See FINRA Letter at 11.

1821 See supra Section II.C.2(c) and Proposing Release, 85 FR at Section III.C.

1822 See FINRA Letter at 9, 11.

1823 See FINRA Letter at 11.

this could prove disruptive to market participants not receiving both information related to OTC equities and core data if a security switches between being listed and unlisted. However, the extent of these effects is uncertain and would depend on the fees FINRA charges for the data.[1824] Market participants may still receive both of these data elements in the same data feed because competing consolidators would be able to offer a product that contains both information related to OTC equities as well consolidated market data.[1825] The degree to which competing consolidators offer this product will depend on the fees FINRA charges for this data as well as the fees for consolidated market data offerings set by the NMS plan.[1826]

(b) Effects of New Round Lot Definition

The final amendments will reduce the number of shares included in the definition of a round lot for NMS stocks for which the prior calendar month's average closing price on the primary listing exchange was greater than $250.00.[1827] Higher priced stocks will be grouped into tiers based on their price and stocks in higher price tiers will have fewer shares in their definition of a round lot. In addition, part of the definition of core data will require that the best bid and offer and national best bid and offer include odd-lots that, when aggregated, are equal to or greater than a round lot and that such aggregation shall occur across multiple prices and shall be

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1824 See supra Section II.C.2(c).

1825 Competing consolidators will not be restricted from also offering data elements from SRO proprietary data. See supra note 220 and accompanying text.

1826 See supra Section II.C.2(c) (discussing fees for information related to OTC equities) and infra Section V.C.2(b)(i) (discussing fees for consolidated market data).

1827 See supra Section II.D.2.

disseminated at the least aggressive price of all such aggregated odd-lots.[1828] Round lot quotes will be protected quotations subject to the trade-through prevention requirements of Rule 611 and the locked and crossed markets restrictions of Rule 610(d).[1829]

For stocks priced above $250, the new round lot definition will result in the inclusion of quotes at better prices in core data that were previously excluded from being reported because they consisted of too few shares. This will make these quotes visible to anyone who subscribes to core data, thereby improving transparency.[1830] This will also mechanically narrow NBBO spreads for most stocks with prices greater than $250, which will affect other Commission or SRO rules and regulations. This section discusses the effects of the new round lot definition on: the NBBO, market participants, the internalization of retail order flow, and on trading venues. Additionally, this section discusses the effects of the monthly calculation to determine the round lot size, the costs of the new round lot definition, and the effects of the new round lot definition on other rules and regulations.

(i) Effects on the NBBO

The new round lot definition will change the average spread between the NBBO for many stocks with prices above $250 because the NBBO will now be calculated based off of the

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1828 See supra Section II.E.2(b). Several exchanges already aggregate odd-lot orders into round lots and report such aggregated odd-lot orders as quotation information to the exclusive SIPs. See Proposing Release, 85 FR at n.157–58.

1829 See supra Section II.E.2(a).

1830 The Vendor Display Rule will require broker-dealers to show, in the context of which a trading or order-routing decision can be implemented, a consolidated display that includes quotes derived from the new round lot size. See supra Section II.D.2(b) and infra Section V.C.1(b)(vii).

smaller round lot size. Because odd-lot shares exist in these stocks at prices that are better than the national best bid and offer (i.e., at prices higher than the national best bid and prices lower than the national best offer), the new national best bid and offer will be at a higher/lower price because fewer odd-lot shares will need to be aggregated together (possibly across multiple price levels) to form a round lot. This will result in a quoted spread that is calculated based off of the NBBO being narrower for these stocks.[1831] The reduction in spreads will be greater in higher priced stocks because stocks in higher priced tiers will have fewer shares included in the definition of a round lot.[1832]

The Commission believes that market participants relying on new core data will see a significant improvement in the NBBO for stocks that fall into the higher priced round lot tiers. Table 3 confirms this belief by updating the analysis from the Proposing Release to account for the new round lot tier structure.[1833] Specifically, Table 3 shows the percentage of instances in a sample of market data when the NBBO provided at the time by an exclusive SIP[1834] was inferior to the price of an NBBO determined by the new definition of a round lot in the final amendments. For instance, the table shows that for stocks with prices between $1,000.01 and $10,000, the new round lot definition caused a quote to be displayed that improved on the current round lot quote 47.7% of the time. The frequency of this narrower NBBO is lower for lower priced stocks. For example, the new round lot definition resulted in a quote being displayed that

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1831 Commenters agreed that the new round lot size would tighten spreads. See, e.g., Nasdaq Letter III at 11, 15; ICI Letter at 6; BestEx Research Letter at 6; CBOE Letter at 5.

1832 See supra Section II.D.2(a).

1833 See Proposing Release, 85 FR at Table 4.

1834 Since the source used for this SIP NBBO is an exclusive SIP itself, this quote includes quotes the exchanges produce by aggregating or "rolling up" odd-lots to obtain a round lot-sized quote.

improved on the current round lot quote 26.6% of the time in the $250.01-$1000 tier. This analysis shows that, within each round lot tier the new round lot definition will improve the quoted spread in a significant number of instances.

Table 3. Instances of Smaller NBBO

Round Lot Tier[1,2]	**Instances of Smaller NBBO (%)[3]**		
	Best Bid	**Best Ask**	**Best Bid or Best Ask**
1. $0-$250 (100 shares)	n/a	n/a	n/a
2. $250.01 - $1,000 (40 shares)	16.3	16.5	26.6
3. $1000.01 - $10,000 (10 shares)	40.2	34.6	47.7
4. $10,000.01+ (1 share)	n/a	n/a	n/a

1. Tier based on the stock's prior month's average closing price in April 2020.

2. Twelve stocks trade in round lots different than 100 shares and are included in the table. Six stocks are in the $0-250 tier and currently trade in 10 share lots, 2 stocks are in the $250.01-$1,000 tier and trade in 10 share lots, 3 stocks are in the $1000.01 to $10,000 tier, and 1 stock is in the $10,000.01+ tier. In the $1000.01 to $10,000 tier, 1 stock trades in 1 share lots and 2 stocks trade in 10 share lots. In the $10,000.01+ tier, 1 stock trades in 1 share lots.

3. Overall frequency of smaller NBBO quotes during May 2020 for the new round lot tier criteria (source: direct feeds) versus the current 100 share round lot criteria (source: SIP). The denominator consists of hourly snapshots from 10:30am to 3:30pm for each trading day in May 2020. The numerator is the total number of snapshots with smaller NBBO quotes.

The effects of instances of narrower NBBOs under the new round lot definition depends on the trading volume of stocks in the tiers affected by the change. The Commission believes that, in particular, for securities with a significant amount of dollar trading volume, there will be significant changes to (tightening of) the quoted spread displayed under the new round lot definition. Table 4 accounts for the new round lot definition, showing the number of NMS stocks that would be in each round lot tier based on monthly average closing prices in September of 2020, as well as the percent of overall average daily volume ("ADV") and notional value

("$ADV") of each price group during one week of trading in October of 2020.[1835] It shows that while most stocks, approximately 98.5%, will remain unaffected by the new round lot definitions, around 28.1% of the dollar trading volume currently is in stocks that will have a new round lot definition.

Table 4. Round Lot Tier Number of Stocks and Trading Volume

Round Lot Tier[1]	Number of Stocks in Round Lot Tier	Percent of ADV, by Price Group[2]	Percent of $ADV, by Price Group[2]
1. $0-$250 (100 shares)	9,023	97.12%	71.93%
2. $250.01 - $1,000 (40 shares)	117	2.79%	23.24%
3. $1000.01 - $10,000 (10 shares)	16	0.09%	4.82%
4. $10,000.01+ (1 share)	1	0.00%	0.02%

1. Tier based on the stock's prior month's average closing price in September 2020.
2. Percent of ADV and Percent of $ADV are based on trading volume between October 5-9, 2020.

The Commission believes that the size of the change in the spread, conditional on the NBBO being smaller, will also be substantial. Table 5 confirms this belief by updating the analysis from the Proposing Release that quantifies the average change in the spread offered by the best quote under the new round lot definition, conditional on the event that the NBBO is smaller in the first place.[1836] The table shows, for example, that the new round lot definition in the $250.01-$1000 tier could yield a 7 basis point reduction in the spread (conditional on the NBBO being smaller). Because the average quoted half spread is 24 basis points, this represents a significant reduction in the half spread. In the case of the $1000.01 to $10,000 tier, the

1835 See Proposing Release, 85 FR at Table 1.

1836 See Proposing Release, 85 FR at Table 5.